

Design a knee prosthesis with a porous titanium coating combined with a hydrogel-based drug delivery system that promotes bone ingrowth while reducing micromotion at the interface

Executive Technology Brief

Domain: mechanical_structural

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'loosening_of_implant_not_related_to_bone' for knee prosthesis: 18 MAUDE reports across 4 distinct devices (20210719 to 20250827); root-cause evidence from 4 recall records and 10 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 4 recall root-cause descriptions (FDA enforcement records, custody attached); 10 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, cust Unresolved gap: Across the 10 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, problem, producing, prosthesis, recurring,

Mechanism: Multi-physics hybrid mechanism combining mechanical fixation and bioactive surface treatment

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Design a knee prosthesis with a porous titanium coating combined with a hydrogel-based drug delivery system that promotes bone ingrowth while reducing micromotion at the interface

Governing Model: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: CONDITIONAL; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value – geometry factors must be applied from the actual design. Engaged variables: NOT ESTABLISHED.

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: CONDITIONAL; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation – use their measured plateau relations instead. Engaged variables: NOT ESTABLISHED.

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: APPLICABLE; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: K.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.